

Lecture 7

List Views and Array Adapter

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ListView

- Android **ListView** is a view which groups several items and display them in vertical scrollable list
- The list items are automatically inserted to the list using an **Adapter** that pulls content from a source such as an array or database

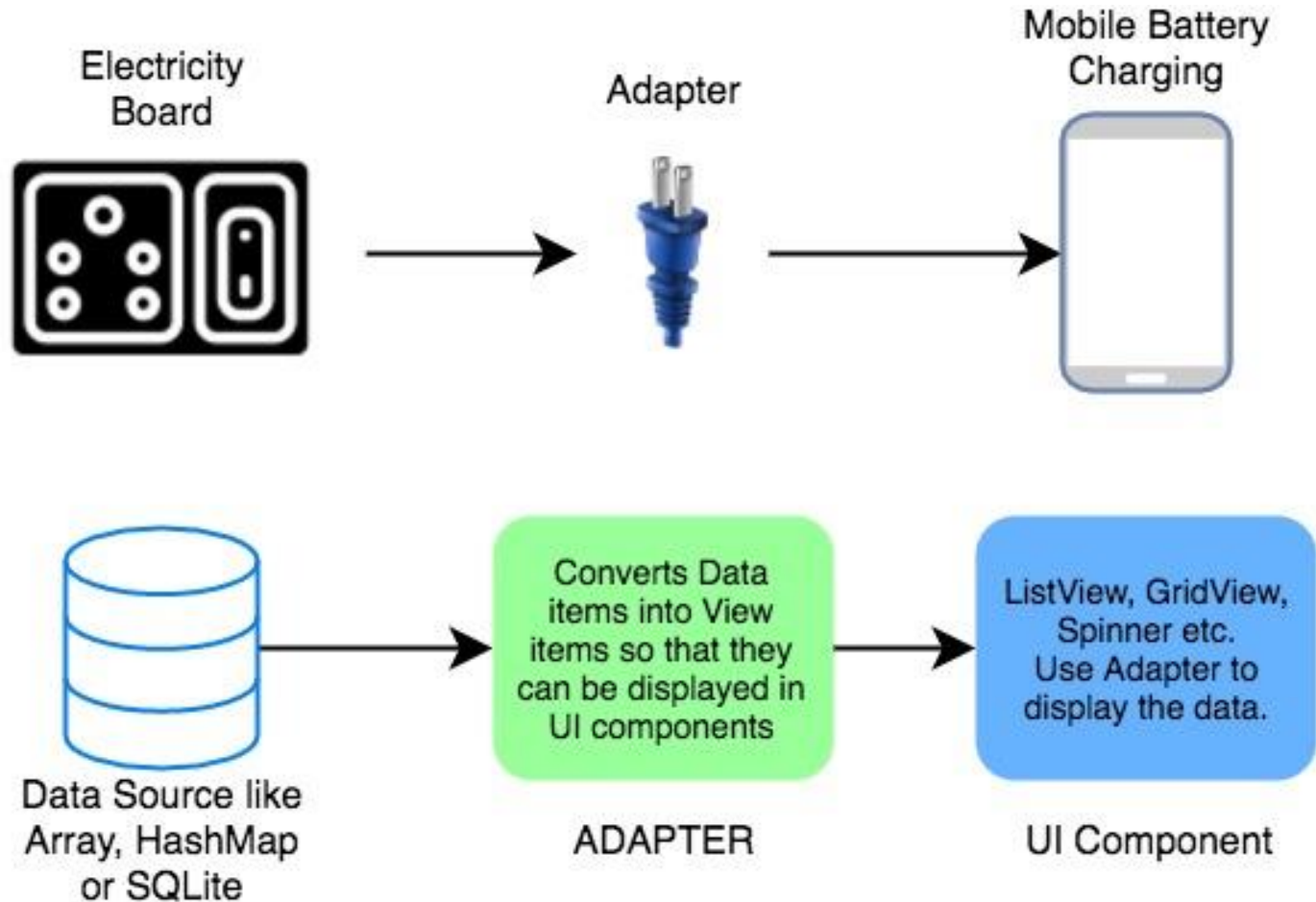
ListView Attributes

- **android:id**
 - This is the ID which uniquely identifies the layout
- **android:divider**
 - This is drawable or color to draw between list items
- **android:dividerHeight**
 - This specifies height of the divider. This could be in px, dp, sp, in, or mm

Adapters

- An adapter actually bridges between UI components and the data source that fill data into UI Component
- Adapter holds the data and send the data to adapter view, the view can takes the data from adapter view and shows the data on different views like as spinner, list view, grid view etc
- The **ListView** is subclasses of **AdapterView** and they can be populated by binding them to an **Adapter**, which retrieves data from an external source and creates a View that represents each data entry

Adapters (Cont.)



Adapters Types

- Android provides several subclasses of Adapter that are useful for retrieving different kinds of data and building views for an AdapterView (i.e. ListView or GridView)
- The common adapters are
 - ArrayAdapter
 - BaseAdapter
 - CursorAdapter
 - SimpleCursorAdapter
 - SpinnerAdapter
 - WrapperListAdapter

← We will cover these two adapters in the class

ArrayAdapter

- You can use this adapter when your data source is an array
- By default, ArrayAdapter creates a view for each array item by calling toString() on each item and placing the contents in a **TextView**
- Consider you have an array of strings you want to display in a ListView, initialize a new **ArrayAdapter** using a constructor to specify the layout for each string and the string array

```
ArrayAdapter adapter = new ArrayAdapter<String>(this, R.layout.list_view_item, stringArray);
```

ArrayAdapter Cont.

- Here are arguments for this constructor –
 - First argument **this** is the application context. Most of the case, keep it **this**
 - Second argument will be layout defined in XML file and having **TextView** for each string in the array
 - Final argument is an array of strings which will be populated in the text view
- Once you have array adapter created, then simply call **setAdapter()** on your **ListView** object as follows

```
ListView listView = (ListView) findViewById(R.id.listview);  
listView.setAdapter(adapter);
```


activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <ListView
        android:id="@+id/mobile_list"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"></ListView>
</LinearLayout>
```

list_view_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<!-- Single List Item Design -->
<TextView

xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/label"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="10dip"
    android:textSize="16dip"
    android:textStyle="bold" >

</TextView>
```

MainActivity.java

```
package com.example.listviewexample;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;

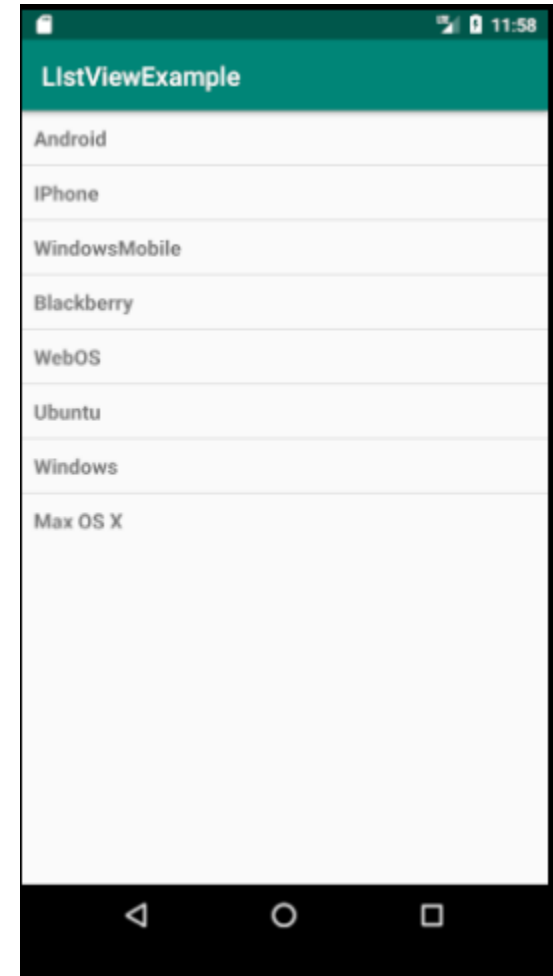
public class MainActivity extends AppCompatActivity {

    // Array of strings...
    String[] mobileArray = {"Android", "iOS", "WindowsMobile", "Blackberry", "WebOS", "Ubuntu",
        "Windows", "Max OS X"};

    ListView operatingSystemList;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ArrayAdapter adapter = new ArrayAdapter<String>(this, R.layout.list_view_layout,
            mobileArray);
        operatingSystemList = (ListView) findViewById(R.id.mobile_list);
        operatingSystemList.setAdapter(adapter);
    }
}
```



Adding List using string.xml

```
<resources>
  <string name="app_name">ListViewExample</string>
  <string-array name="os_list">
    <item>Android</item>
    <item>iOS</item>
    <item>WindowsMobile</item>
    <item>Blackberry</item>
    <item>WebOS</item>
    <item>Ubuntu</item>
    <item>Windows</item>
    <item>Max OS X</item>
  </string-array>
</resources>
```

MainActivity.java

```
package com.example.listviewexample;

import androidx.appcompat.app.AppCompatActivity;

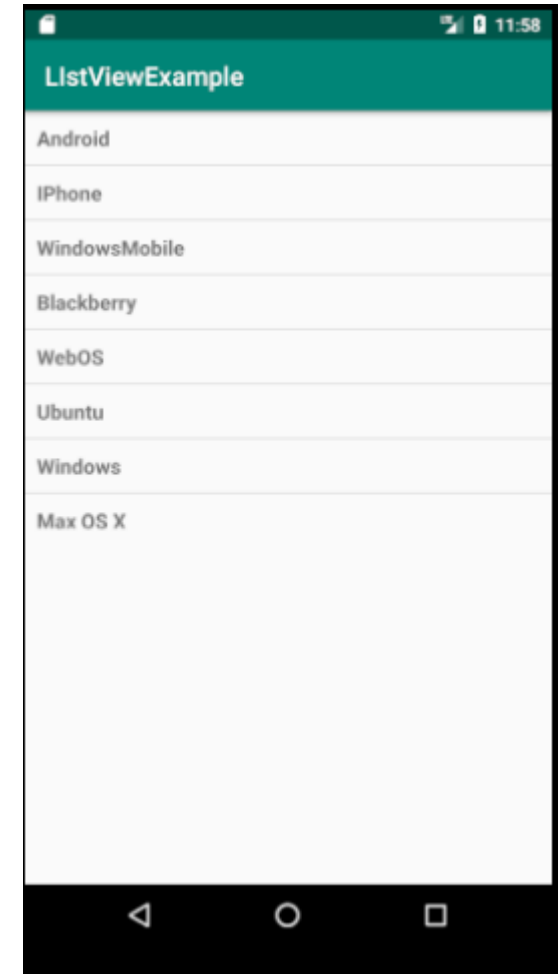
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;

public class MainActivity extends AppCompatActivity {

    ListView operatingSystemList;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ArrayAdapter adapter = new ArrayAdapter<String>(this, R.layout.list_view_item,
            getResources().getStringArray(R.array.os_list));
        operatingSystemList = (ListView) findViewById(R.id.mobile_list);
        operatingSystemList.setAdapter(adapter);
    }
}
```



List View Item Click Listener

- To add list view item click listener, `OnItemClickListener` can be used

```
// list view click listener
operatingSystemList.setOnItemClickListener(new AdapterView.OnItemClickListener() {
    @Override
    public void onItemClick(AdapterView<?> parent, View view, int position, long id) {

        // perform any task on click listener
        Toast.makeText(getApplicationContext(), mobileArray[position], Toast.LENGTH_SHORT).show();
    }
});
```

Update List View on Run Time using InvalidateViews()

```
public class MainActivity extends AppCompatActivity {

    // Array of strings...
    ArrayList<String> mobileArray = new ArrayList<String>();

    ListView operatingSystemList;
    EditText addNewItem;
    Button updateItem;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        mobileArray.add("Android");
        mobileArray.add("iOS");
        mobileArray.add("WindowsMobile");
        mobileArray.add("WebOS");

        ArrayAdapter adapter = new ArrayAdapter<String>(this, R.layout.list_view_layout,
            mobileArray);

        operatingSystemList = (ListView) findViewById(R.id.mobile_list);
        addNewItem = (EditText) findViewById(R.id.add_new_item_tv);
        updateItem = (Button) findViewById(R.id.update_item_btn);
        operatingSystemList.setAdapter(adapter);

        // list view click listener
        operatingSystemList.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
                Toast.makeText(getApplicationContext(), mobileArray.get(position), Toast.LENGTH_SHORT)
                    .show();
            }
        });

        public void onClickUpdateItem(View view){
            String newItem = addNewItem.getText().toString();
            if(newItem.isEmpty()) {
                mobileArray.add(newItem);
                operatingSystemList.invalidateViews();
            }
        }
    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
tools:context=".MainActivity">

    <EditText
        android:id="@+id/add_new_item_tv"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName"
        android:text="Name" />

    <Button
        android:id="@+id/update_item_btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:onClick="onClickUpdateItem"
        android:text="Update" />

    <ListView
        android:id="@+id/mobile_list"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toTopOf="parent"></ListView>

</LinearLayout>
```

